

Program: Center:

1



Let's Explore!

Mike is in a tutoring session. He needs to know how compilers work.



Instructions: Read the text below.

Mike: What do you need to do when you want to transform the source code into a computer language?

Tutor: You need to use a compiler. This is a program that can process data from a source language to translate it into a target language. The output or result of the compilation process is an object code that can be read by a processor.

Mike: So, when you want to create an executable file, do you have to use a compiler?

Tutor: That's right. A compiler creates the encoded instructions of an executable.

Mike: How does a compiler work?

Tutor: Do you want to know how it works? ... Look at this diagram! When a compiler is running, it makes a lexical analysis, a parsing and a semantic analysis of the source language statements. The compiler generates an intermediate code and it seeks to match the information correctly to build and optimize the output code.

Mike: Are all compilers similar?

Tutor: No, there are different kinds of compilers. Programmers use most compilers to translate a high level programming language to a lower level. However, there are some that have different features.

Mike: For instance? Do you want to give me some examples?

Tutor: For example, decompilers, these programs can do the opposite process; they translate from a low level to a more difficult level of computer language. We have also cross-compilers; these kinds of compilers are able to run in different types of operating systems; it doesn't matter if the compiler runs in a different system from the one where the executable runs.

Mike: Are there source-to-source compilers?

Tutor: Yes, they are known as transpilers too and they can translate source code with the same high level programming language.



**Let's Practice!**

Do you know what a compiler is?

*Instructions: Complete the sentences by writing its corresponding word. Use capital letter where necessary. Use: **need to, translate, is, can process, need to use***

Mike: What do you _____ do when you want to transform source code language into another computer language?

Tutor: You _____ a compiler. This is a program that _____ data from a source language to _____ it into a target language. The output or result of the compilation process _____ an object code that can be read by a processor.

**Let's Practice!**

Let's see some operations that compilers can perform.

Instructions: Proofread the sentences. Correct the mistakes.

1. A compiler create the encoded instructions of an executable.

2. When a compiler is running, it make a lexical analysis, a parsing and a semantic analysis of the source language statements.

3. The compiler generate an intermediate code and it seeks to match the information correctly to build and optimize the output code.

4. Programmers uses most compilers to translate a high level programming language to a lower level.





Extension Activity

Let's compare two types of compilers.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Choose two types of compilers.
- Look for detailed information about them.
- Make a comparative chart with the information you found.
- Explain your chart to your instructor.

Self-Assessment

Yes

No

- I choose two types of compilers.
- I make a comparative chart.
- I explain my chart to my instructor.

☐	☐
☐	☐
☐	☐



Target Words

- **parsing** (verb-present participle): refers to a syntactic analysis of a computer language.
e.g. When a compiler is running, it makes a lexical analysis, a parsing and a semantic analysis of the source language statements.
- **output code** (adjective phrase): the resulting code after a compilation process.
e.g. The compiler generates an intermediate code and it seeks to match the information correctly to build and optimize the output code.
- **run** (verb): to execute an operation.
e.g. if doesn't matter if the compiler runs in a different system from the one where the executable runs.
- **target** (adjective): relating to a goal.
e.g. This is a program that can process data from a source language to translate it into a target language.
- **executable file** (adjective phrase): a file that contains encoded data.
e.g. So, when you want to create an executable file, do you have to use a compiler?
- **encoded** (adjective): containing encrypted information.
e.g. A compiler creates the encoded instructions of an executable.



Learning Points

Verbs with Infinitives in Questions

In English, there are some *verbs* that are always followed by other verbs in the infinitive form, for example, the verb *want*.

These verbs are always followed by the preposition *to* and they can express different meanings. When you use the infinitive verb *want to*, it refers to the desire or the aim of doing or getting something:

For example:

I **want to** learn how to use a compiler.

Do you **want to** learn how to use a compiler?

There are some other meanings when using other *infinitive verbs* such as:

Verbs that express thinking or feeling: *decide, choose, hope, like, prefer, expect, etc.* .

Verbs of speaking: *promise, agree, refuse, etc.*

Other verbs: *want to, have, help, want, try, need, manage, plan, need, etc.*

You can use *verbs and modal verbs followed by infinitives in questions*.

For instance: Do you **want to** transform source code from a program into another computer language?

What do you **want to** use a standard or a source to code compiler?

Do you **have to** look for a compiler?

What do you **need to** know to run a compiler?

Look at the chart to see how to build questions using *verbs with infinitives*:

WH WORD	AUXILIARY VERB	SUBJECT	VERB WITH INFINITIVE	COMPLEMENT
	<i>Do</i>	<i>you</i>	<i>want to</i>	<i>learn how to use a compiler?</i>
<i>What</i>	<i>do</i>	<i>you</i>	<i>want to</i>	<i>use a standard or a source to code compiler?</i>

Examples

- *Do you want to know how it works?*
- *Do you want to give me some examples?*





Learning Points

Diagram of Operation of a Typical Compiler

Compilers in general use to translate codes from high level programming language to a lower level.

The operation of a typical compiler goes through some important stages:

1. The source code language passes first by a lexical, syntactical (parsing) and semantical analysis.
2. Then, the compiler generates an intermediate code or language.
3. After this, the code goes to an intermediate code optimizer.
4. Once the intermediate code passes the error checking and the data is validated, it ends up in the generation of the object code.

Look at the diagram to identify the stages of the compilation process:

Example

- *Look at this diagram! When a compiler is running, it makes a lexical analysis, a parsing and a semantic analysis of the source language statements. The compiler generates an intermediate code and it seeks to match the information correctly to build and optimize the output code.*